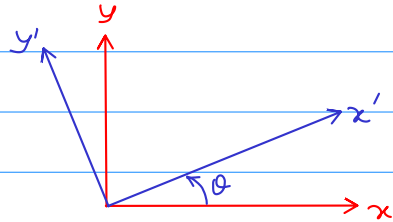


Stress Transformation in 3D

Recap of 2D transformation



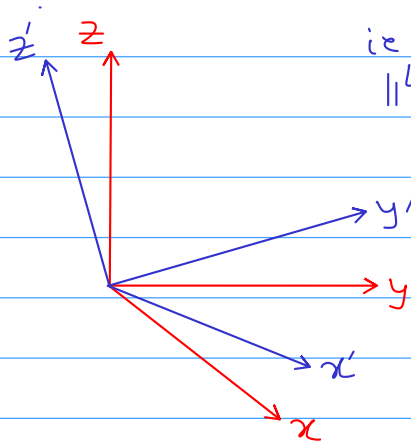
$$L = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

dc's of $\hat{e}_{x'}$ = $\langle l_{11}, l_{12} \rangle = \langle \cos\theta, \sin\theta \rangle$

dc's of $\hat{e}_{y'}$ = $\langle l_{21}, l_{22} \rangle = \langle \cos(90+\theta), \sin(90+\theta) \rangle = \langle -\sin\theta, \cos\theta \rangle$

$$\Rightarrow L = \begin{bmatrix} l_{11} & l_{12} \\ l_{21} & l_{22} \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

In, 3D:- Let dc's of $\hat{e}_{x'}$ relative to (x, y, z) -system
be $l_{11} \ l_{12} \ l_{13}$.



$$\left. \begin{aligned} \text{ie } \hat{e}_{x'} &= \langle l_{11} \ l_{12} \ l_{13} \rangle \\ \text{by } \hat{e}_{y'} &= \langle l_{21} \ l_{22} \ l_{23} \rangle \\ \hat{e}_{z'} &= \langle l_{31} \ l_{32} \ l_{33} \rangle \end{aligned} \right\} \text{--- (98)}$$

Now Consider the plane whose unit normal is $\hat{e}_{x'}$.

we know $p(\hat{n}) = \tau \hat{n}$
replace $\hat{n}$ with $\hat{e}_{x'}$

$$\Rightarrow p(\hat{e}_{x'}) = \tau \hat{e}_{x'}$$

Now,

$$\begin{aligned} \tau_{x'x'} &= \text{Component of } p(\hat{e}_{x'}) \text{ in } \hat{e}_{x'} \text{ direction} \\ &= \text{dot product of } \hat{e}_{x'} \text{ \& } p(\hat{e}_{x'}) \end{aligned}$$

$$\Rightarrow \tau_{xx} = \hat{e}_x \cdot p(\hat{e}_x)$$

$$= [\hat{e}_x]^T p(\hat{e}_x)$$

$$= [\hat{e}_x]^T \tau \hat{e}_x$$

$$= \begin{bmatrix} l_{11} & l_{12} & l_{13} \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} \begin{bmatrix} l_{11} \\ l_{12} \\ l_{13} \end{bmatrix}$$

$$= l_{11}^2 \tau_{xx} + l_{11} l_{12} \tau_{xy} + l_{11} l_{13} \tau_{xz} + l_{12} l_{11} \tau_{yx} + l_{12}^2 \tau_{yy} + l_{12} l_{13} \tau_{yz} + l_{13} l_{11} \tau_{zx} + l_{13}^2 \tau_{zz} \quad \text{--- (99)}$$

The other eight components of the stress tensor in the primed co-ordinate system can be found in a similar manner

$$\text{eg } \tau_{x'y'} = \hat{e}_{y'} \cdot p(\hat{e}_{x'}) \text{ etc.}$$

The resulting Eqns in the form of Eqn 99 can be written in matrix form as under:

$$\begin{bmatrix} \tau_{x'x'} & \tau_{x'y'} & \tau_{x'z'} \\ \tau_{y'x'} & \tau_{y'y'} & \tau_{y'z'} \\ \tau_{z'x'} & \tau_{z'y'} & \tau_{z'z'} \end{bmatrix} = \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ l_{21} & l_{22} & l_{23} \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ l_{21} & l_{22} & l_{23} \\ l_{31} & l_{32} & l_{33} \end{bmatrix}^T$$

→ (100)

or, Symbolically,

$$\boxed{\tau' = L \tau L^T} \quad \text{--- (101)}$$

Note:- Here, the rotation matrix has been formed using the d.c's of three primed unit vectors.

Using the Symmetry properties of the Stress Tensor
Eqn 99 can be simplified to.

$$\tau_{x'x'} = l_{11}^2 \tau_{xx} + l_{22}^2 \tau_{yy} + l_{33}^2 \tau_{zz} + 2l_{11}l_{12} \tau_{xy} + 2l_{11}l_{13} \tau_{xz} + 2l_{12}l_{13} \tau_{yz} \quad (102)$$

Similarly other 5-component can be written as.

$$\tau_{y'y'} = l_{21}^2 \tau_{xx} + l_{22}^2 \tau_{yy} + l_{23}^2 \tau_{zz} + 2l_{21}l_{22} \tau_{xy} + 2l_{21}l_{23} \tau_{xz} + 2l_{22}l_{23} \tau_{yz} \quad (103)$$

$$\tau_{z'z'} = l_{31}^2 \tau_{xx} + l_{32}^2 \tau_{yy} + l_{33}^2 \tau_{zz} + 2l_{31}l_{32} \tau_{xy} + 2l_{31}l_{33} \tau_{xz} + 2l_{32}l_{33} \tau_{yz} \quad (104)$$

$$\tau_{x'y'} = \left. \begin{aligned} & l_{11}l_{21} \tau_{xx} + l_{12}l_{22} \tau_{yy} + l_{13}l_{23} \tau_{zz} + (l_{11}l_{22} + l_{12}l_{21}) \tau_{xy} \\ & + (l_{12}l_{23} + l_{13}l_{22}) \tau_{yz} + (l_{11}l_{23} + l_{13}l_{21}) \tau_{xz} \end{aligned} \right\} \quad (105)$$

$$\tau_{y'z'} = \left. \begin{aligned} & l_{21}l_{31} \tau_{xx} + l_{22}l_{32} \tau_{yy} + l_{23}l_{33} \tau_{zz} + (l_{21}l_{32} + l_{22}l_{31}) \tau_{xy} \\ & + (l_{22}l_{33} + l_{23}l_{32}) \tau_{yz} + (l_{21}l_{33} + l_{23}l_{31}) \tau_{xz} \end{aligned} \right\} \quad (106)$$

$$\tau_{x'z'} = \left. \begin{aligned} & l_{11}l_{31} \tau_{xx} + l_{12}l_{32} \tau_{yy} + l_{13}l_{33} \tau_{zz} + (l_{11}l_{32} + l_{12}l_{31}) \tau_{xy} \\ & + (l_{12}l_{33} + l_{13}l_{32}) \tau_{yz} + (l_{11}l_{33} + l_{13}l_{31}) \tau_{xz} \end{aligned} \right\} \quad (107)$$

If the unprimed Co-ordinate System is the Co-ordinate System of Principal Stresses

Then, Eqn (102) to (107) takes the following form.

$$\tau_{x'x'} = l_{11}^2 \tau_{xx} + l_{22}^2 \tau_{yy} + l_{33}^2 \tau_{zz} \quad (108)$$

$$\tau_{y'y'} = l_{21}^2 \tau_{xx} + l_{22}^2 \tau_{yy} + l_{23}^2 \tau_{zz} \quad (109)$$

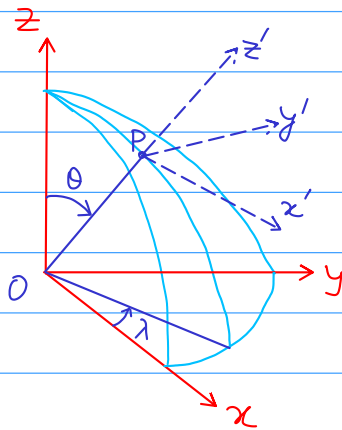
$$\tau_{z'z'} = l_{31}^2 \tau_{xx} + l_{32}^2 \tau_{yy} + l_{33}^2 \tau_{zz} \quad (110)$$

$$T_{x'y'} = l_{11}l_{21}T_{xx} + l_{12}l_{22}T_{yy} + l_{13}l_{23}T_{zz} \quad \text{--- (111)}$$

$$T_{y'z'} = l_{21}l_{31}T_{xx} + l_{22}l_{32}T_{yy} + l_{23}l_{33}T_{zz} \quad \text{--- (112)}$$

$$T_{x'z'} = l_{11}l_{31}T_{xx} + l_{12}l_{32}T_{yy} + l_{13}l_{33}T_{zz} \quad \text{--- (113)}$$

Other System



Sometimes it is convenient to use λ & θ to specify a plane, rather than $dc's$ of the plane relative to Cartesian Co-ordinate System.

where,

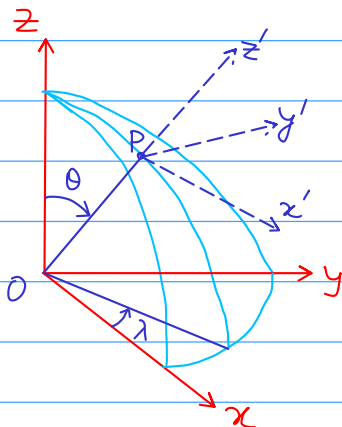
λ = longitude angle.

θ = zenith angle

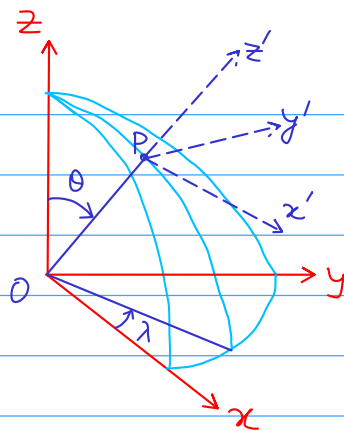
Axes associated with the primed Co-ordinate System

- (i) $P_{z'}$ → radial direction
- (ii) $P_{x'}$ → lies in plane OPz & associated with angle θ .
- (iii) $P_{y'}$ → Chosen in such a way so as to complete Right Handed coordinate system & points in the direction increasing λ .

The components of these unit Vectors relative to unprimed Co-ordinate System is given by Eqn (98) expressed in terms of θ & λ .



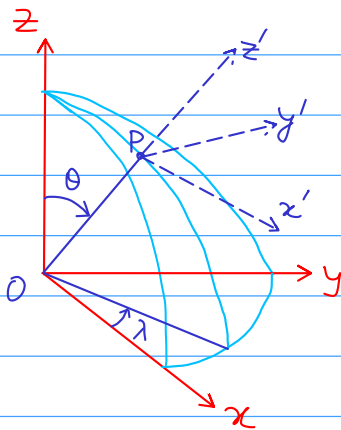
$$\hat{e}_{z'} = (\underbrace{\sin\theta\cos\lambda}_{\substack{\uparrow \\ \text{dc wrt} \\ x\text{-axis}}}, \underbrace{\sin\theta\sin\lambda}_{\substack{\downarrow \\ \text{dc wrt} \\ y\text{-axis}}}, \underbrace{\cos\theta}_{\substack{\downarrow \\ \text{dc wrt} \\ z\text{-axis}}}) \quad \text{--- (114)}$$



Since,
 $P_{x'}$ $\rightarrow$ lies in plane OPz & associated with angle θ .

$\therefore$ The Component of Unit vector through O parallel to $P_{x'}$ can be found by replacing $\theta \rightarrow \cos(\theta + \frac{\pi}{2})$ in previous Eqn (114).

$$\therefore \hat{e}_{x'} = (\cos\theta \cos\lambda, \cos\theta \sin\lambda, -\sin\theta) \quad \text{--- (115)}$$



Finally, a unit Vector through O parallel to $P_{y'}$ is $\perp$ to Oz .

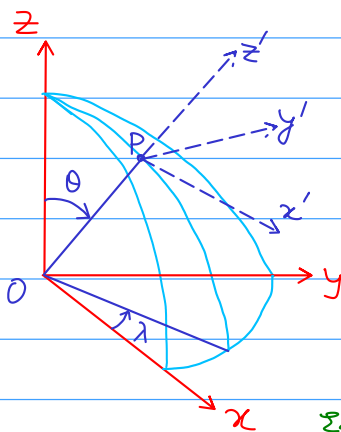
And make an angle $(\lambda + \frac{\pi}{2})$ with Ox

$$\therefore \hat{e}_{y'} = \langle -\sin\lambda, \cos\lambda, 0 \rangle$$

The rotation matrix L is therefore

$$L = \begin{bmatrix} \cos\theta \cos\lambda & \sin\lambda \cos\theta & -\sin\theta \\ -\sin\lambda & \cos\lambda & 0 \\ \sin\theta \cos\lambda & \sin\theta \sin\lambda & \cos\theta \end{bmatrix}$$

Example



Suppose we are interested to know the Components of traction on $Px'y'$ plane. Given (x, y, z) is the principal Co-ordinate system.

The components in this plane will be $\tau_{z'z'}, \tau_{z'y'}, \tau_{z'x'}$

$\tau' = L \tau L^T$ can be used if Numerical values are given.

However, here we use the direct formula already derived

Eqn (110) gives

$$\tau_{z'z'} = l_{31}^2 \tau_{xx} + l_{32}^2 \tau_{yy} + l_{33}^2 \tau_{zz}$$

Since (x, y, z) is principal coordinate system

$$\therefore \tau_{xx} = \sigma_1, \tau_{yy} = \sigma_2 \quad \& \quad \tau_{zz} = \sigma_3.$$

Now from above Rotation matrix

$$l_{31} = \sin\theta \cos\lambda, \quad l_{32} = \sin\theta \sin\lambda \quad \& \quad l_{33} = \cos\theta.$$

$$\therefore \tau_{z'z'} = (\sin\theta \cos\lambda)^2 \sigma_1 + (\sin\theta \sin\lambda)^2 \sigma_2 + (\cos^2\theta) \sigma_3$$

$$\tau_{z'z'} = [\sigma_1 \cos^2\lambda + \sigma_2 \sin^2\lambda] \sin\theta + \sigma_3 \cos^2\theta. \quad \text{--- (117)}$$

With a similar approach.

$$\tau_{z'y'} = -\frac{1}{2} (\sigma_1 - \sigma_2) \sin\theta \sin 2\lambda \quad \text{--- (118)}$$

$$\tau_{z'x'} = \frac{1}{2} [\sigma_1 \cos^2\lambda + \sigma_2 \sin^2\lambda - \sigma_3] \sin 2\theta \quad \text{--- (119)}$$

Useful for analysis of Tectonic Regimes.

Attendance - 27